

SUPPLEMENTARY MATERIALS: STABLE, HIGH-ORDER EXPONENTIAL IMEX LINEAR MULTISTEP METHODS

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Equation, section, and figure numbers without the prefix SM refer to the main paper, whereas those carrying this prefix refer to the supplement; the notation of the main paper is used throughout.

SM1. Structured multilinear convolution operators.

SM1.1. Translation-equivariant polynomial operators. Consider a periodic grid $x_i = i\Delta x$, $i \in \mathbb{Z}_M$, and let $(T_a v)_i = v_{i-a}$. A symmetric d -linear map $\mathcal{G}_d : (\mathbb{R}^M)^d \rightarrow \mathbb{R}^M$ is translation equivariant when

$$(SM1.1) \quad \mathcal{G}_d[T_a v_1, \dots, T_a v_d] = T_a \mathcal{G}_d[v_1, \dots, v_d] \quad \forall a \in \mathbb{Z}_M.$$

Such a map admits the multilinear convolution representation

$$(SM1.2) \quad (\mathcal{G}_d[v_1, \dots, v_d])_i = \sum_{\ell_1, \dots, \ell_d \in \mathbb{Z}_M} g_d(\ell_1, \dots, \ell_d) (v_1)_{i-\ell_1} \cdots (v_d)_{i-\ell_d},$$

Here g_d is symmetric. We impose locality by setting $g_d = 0$ whenever $\max_r |\ell_r|$ exceeds a prescribed radius. This representation follows from the standard characterization of shift-commuting linear maps applied to each argument [SM1, SM2, SM3]. In Fourier variables,

$$(SM1.3) \quad \mathcal{G}_d[\widehat{v_1}, \dots, \widehat{v_d}](k) = \sum_{k_1 + \dots + k_d = k} \hat{g}_d(k_1, \dots, k_d) \hat{v}_1(k_1) \cdots \hat{v}_d(k_d).$$

Assume that N is a translation-equivariant polynomial of degree at most $d_{\max}$,

$$(SM1.4) \quad N(u) = \sum_{d=0}^{d_{\max}} P_d(u, \dots, u), \quad P_d \text{ symmetric and } d\text{-linear},$$

with local kernels. Its state derivatives therefore belong to the class (SM1.2). Replacing each derivative by a trainable $\mathcal{G}_d$ in the Faà di Bruno formula for $J^{(j)}$ (main paper, Section 2.2) gives

$$(SM1.5) \quad \boxed{\hat{J}^{(j)} v = \sum_{\substack{\Pi \vdash \{1, \dots, j\} \\ |\Pi|+1 \leq d_{\max}}} \mathcal{G}_{|\Pi|+1} [\hat{u}^{(|\beta_1|)}, \dots, \hat{u}^{(|\beta_{|\Pi|}|)}, v], \quad 0 \leq j < S.}$$

The prediction (4.11) is consequently linear in the kernel coefficients and may be calibrated by linear or regularized least squares.

SM1.2. Approximation error. Let

$$(SM1.6) \quad \delta \mathcal{G}_d := \mathcal{G}_d - N^{(d)}(u_n), \quad d \leq d_{\max}.$$

Subtracting the Faà di Bruno formula for $J^{(j)}$ (main paper, Section 2.2) from (SM1.5) produces three error components: tensor approximation, slot reconstruction, and degree truncation. Insert this decomposition into (4.17) and regroup the partitions of

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$\{1, \dots, S\}$ to obtain

$$(SM1.7) \quad \begin{aligned} \varepsilon_S^N = & \sum_{\substack{\sigma \vdash [S] \\ |\sigma| \leq d_{\max}}} \delta \mathcal{G}_{|\sigma|} \left[u^{(|\beta_1|)}, \dots, u^{(|\beta_{|\sigma|}|)} \right] \\ & - \sum_{\substack{\sigma \vdash [S] \\ |\sigma| > d_{\max}}} N^{(|\sigma|)}(u) \left[u^{(|\beta_1|)}, \dots, u^{(|\beta_{|\sigma|}|)} \right] \\ & + \Delta T K_{S-1} \left[\hat{j}^{(0)} u^{(S+1)} + (S-1) \hat{j}^{(1)} u^{(S)} + \mathcal{G}_2[u^{(S)}, u^{(1)}] \right] + \mathcal{O}(\Delta T^2). \end{aligned}$$

The first line is the operator-approximation error, the second is the degree-truncation error, and the third is the finite-difference input error. Under (SM1.4), the truncation term vanishes. If the state derivatives are represented exactly, then

$$(SM1.8) \quad \varepsilon_S^N = \Delta T K_{S-1} \left[N'(u) u^{(S+1)} + S N''(u) [u^{(1)}, u^{(S)}] \right] + \mathcal{O}(\Delta T^2).$$

Thus the unresolved rung remains first-order accurate even when the operator class is exact. This residual is the input term already included in the order analysis of Section 4.3 of the main paper.

SM1.3. Quadratic convolution operators. For $N(u) = P(u, u) + F$, with P symmetric and bilinear, set $\mathcal{G}_2 = 2\hat{P}$ and $\mathcal{G}_1 = 2\hat{P}(u_n, \cdot)$. Then

$$(SM1.9) \quad \hat{j}^{(j)} v = 2\hat{P}(\hat{u}^{(j)}, v), \quad D^{(j)} = 2(\hat{P} - P)(u^{(j)}, \cdot) + 2\hat{P}(\varepsilon_j^u, \cdot),$$

and (SM1.7) reduces to

$$(SM1.10) \quad \begin{aligned} \varepsilon_S^N = & \sum_{r=0}^S \binom{S}{r} (\hat{P} - P)(u^{(r)}, u^{(S-r)}) \\ & + 2K_{S-1} \Delta T \left[\hat{P}(u, u^{(S+1)}) + S\hat{P}(u^{(1)}, u^{(S)}) \right] + \mathcal{O}(\Delta T^2). \end{aligned}$$

This expression separates the bilinear-kernel error from the history reconstruction error and requires only one learned convolution kernel.

SM1.4. Linear convolution operators and Wiener optimality. The restricted class $d_{\max} = 1$ uses a single linear convolution $\mathcal{K}$ and predicts $\hat{N}^{(S)} = \mathcal{K}\hat{u}^{(S)}$. With

$$(SM1.11) \quad b_S := N^{(S)} - N'(u)u^{(S)},$$

the prediction error is

$$(SM1.12) \quad \varepsilon_S^N = (\mathcal{K} - N'(u))u^{(S)} - b_S + \Delta T K_{S-1} \mathcal{K} u^{(S+1)} + \mathcal{O}(\Delta T^2).$$

Let $\chi = \hat{u}^{(S)}$ and $y = N^{(S)}$. The mean-square optimal operator in the admissible convolution space $\mathcal{C}$ minimizes $\mathbb{E}\|y - \mathcal{K}\chi\|^2$ and satisfies the Wiener normal equations

$$(SM1.13) \quad \mathbb{E}\langle y - \mathcal{K}^* \chi, \mathcal{V} \chi \rangle = 0 \quad \forall \mathcal{V} \in \mathcal{C}, \quad \mathcal{K}^* = R_{y\chi} R_{\chi\chi}^{-1}.$$

Consequently, with $\varrho = y - \mathcal{K}^* \chi$,

$$(SM1.14) \quad \mathbb{E}\|y - \mathcal{K}\chi\|^2 = \mathbb{E}\|\varrho\|^2 + \mathbb{E}\|(\mathcal{K} - \mathcal{K}^*)\chi\|^2, \quad \mathcal{K} \in \mathcal{C}.$$

The first term is the irreducible error of the linear convolution class; the second quantifies estimation or optimization error [SM4, SM5].

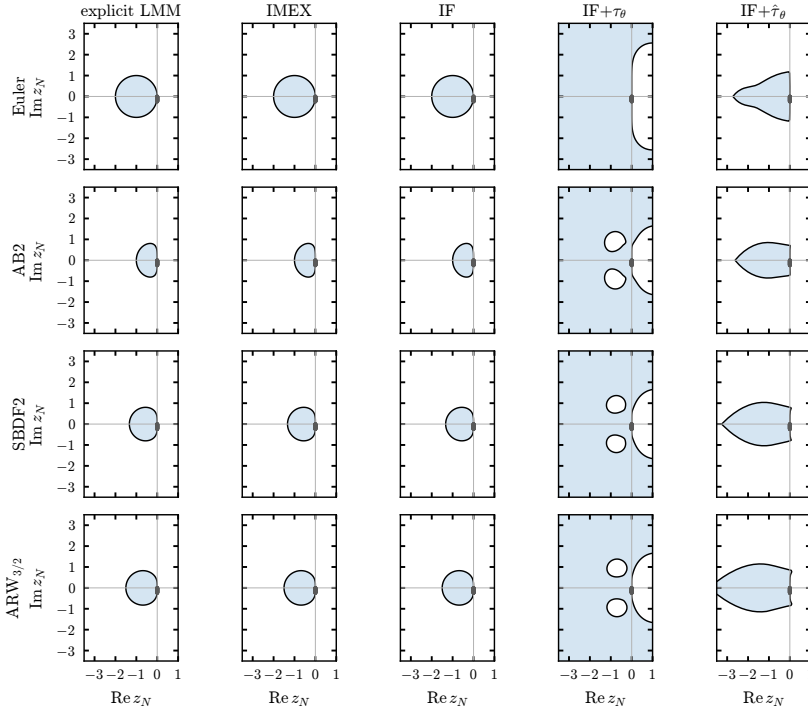
SM2. Modewise stability limits. Table SM2.1 lists the largest $|z_{N,0}|$ on the imaginary and negative real axes for which every companion-polynomial root satisfies the stated tolerance. The table covers the instances in Section 5 of the main paper. It compares explicit and implicitly weighted corrections with exact and learned rungs.

IF scheme	base	exact rungs		learned (multilinear)	
		explicit $\theta = 1$	$\theta = \frac{1}{4}$	explicit $\theta = 1$	$\theta = \frac{1}{4}$
Euler	0 / 2.0	2.8 / 2.8	∞ / ∞^*	0.43 / 0.35	0.54 / 2.7
CNLF ($\gamma = 0$)	1.0 / 0	2.8 / 0	0.53 / 0	0 / 0	0 / 0
ARW ($\gamma = \frac{1}{4}$)	0.30 / 0.67	2.2 / 0.61	$\infty / 0.76$	0.32 / 0.23	0.62 / 2.3
AB2 ($\gamma = \frac{1}{2}$)	0.24 / 1.0	1.7 / 0.85	∞ / ∞	0.43 / 0.35	0.73 / 2.6
SBDF2 ($\gamma = 1$)	0.18 / 1.33	1.3 / 1.0	∞ / ∞	0.54 / 0.47	0.82 / 3.3
ARW ($\gamma = \frac{3}{2}$)	0.15 / 1.5	1.1 / 1.1	∞ / ∞	0.59 / 0.54	0.85 / 3.9

TABLE SM2.1

Largest stable $|z_{N,0}|$ (imaginary axis / negative real axis, $z_L = 0$, $q = S = 4$, tolerance 10^{-3}). ∞ : no crossing up to $|z_{N,0}| = 12$; *except isolated poles of (3.43). For the learned scheme the imaginary-axis entries with $\theta = \frac{1}{4}$ are set by the principal root (leak), the parasitic roots being stable up to 1.2 (Euler) and 0.73 (AB2).

SM3. Additional figures. Figures SM3.1–SM3.10 complement Section 5 of the main paper. They include stability regions for the KdV and Burgers testbeds, including the raw surrogate stencil. They also report amplification, convergence, cost, and error histories for the three testbeds.


 FIG. SM3.1. KdV testbed, stability regions at $z_L = 0$; layout as in Figure 5.1 of the main paper.

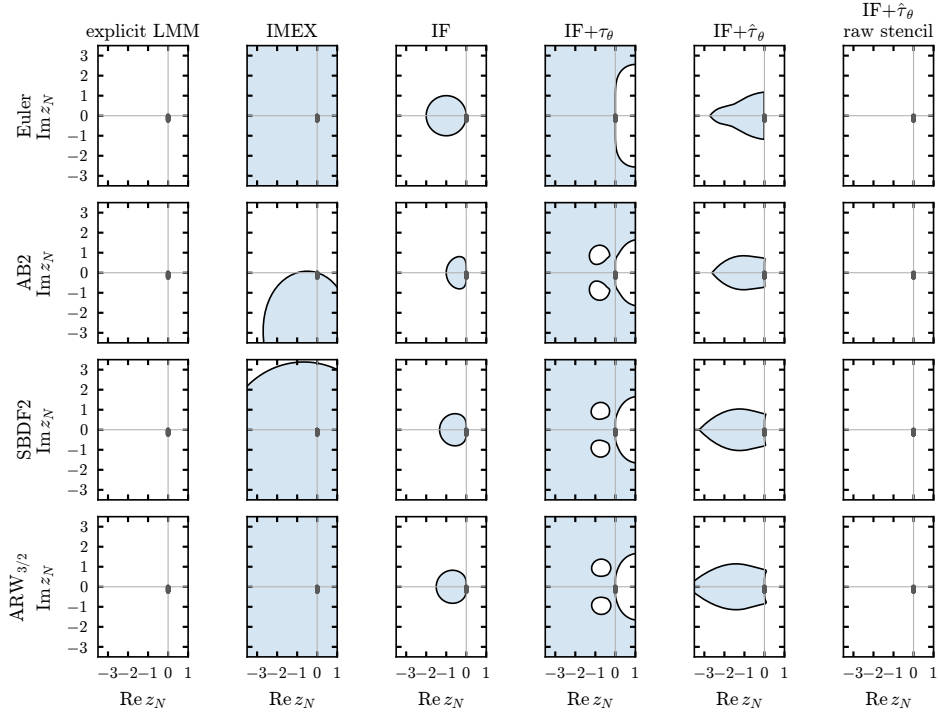


FIG. SM3.2. *KdV* tested at $z_L = 10i$ with a sixth column: the surrogate reading the raw stencil (untransported histories). With the raw stencil the finite-difference taps carry the factors $(-z_L)^j$ and E^{1-j} and the learned correction has no stability region at $|z_L| = 10$.

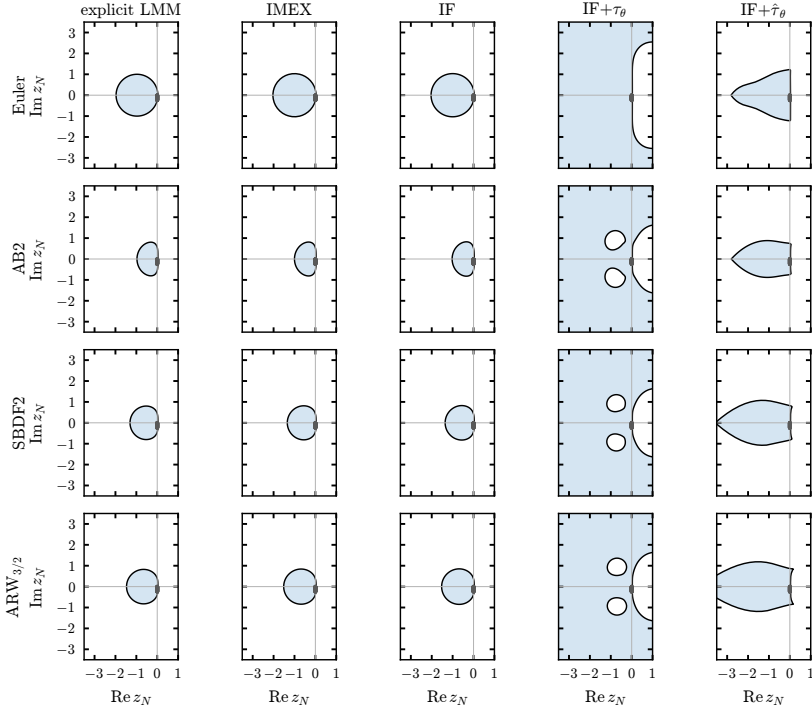


FIG. SM3.3. *Burgers testbed, stability regions at the mode of largest $|z_L|$ ($z_L = -0.03$).*

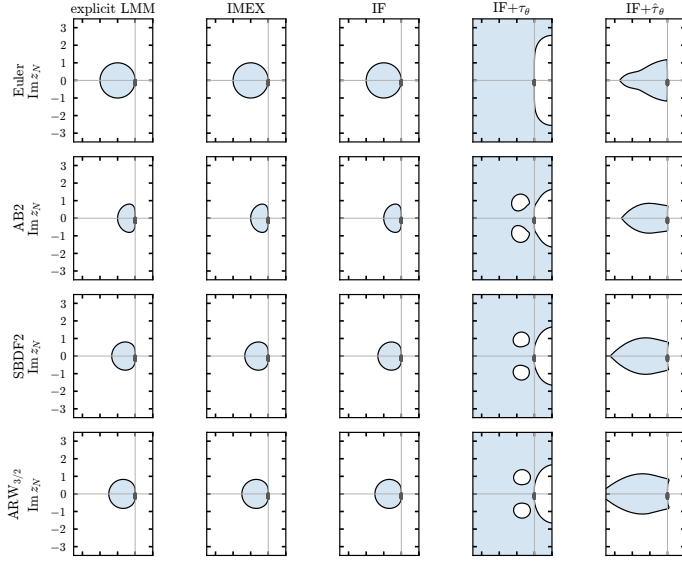


FIG. SM3.4. *Burgers testbed, stability regions at $z_L = 0$.*

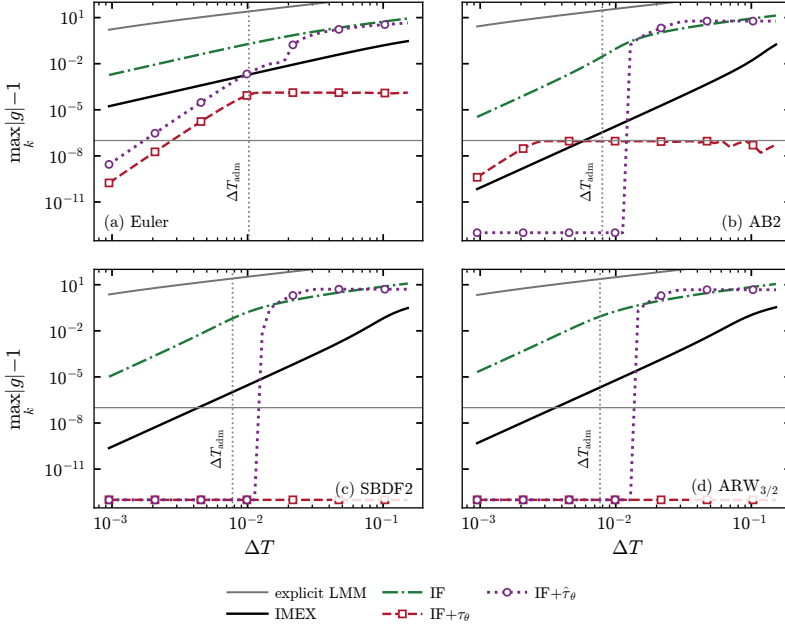


FIG. SM3.5. *KdV testbed: largest excess amplification versus ΔT ; layout as in Figure 5.3 of the main paper.*

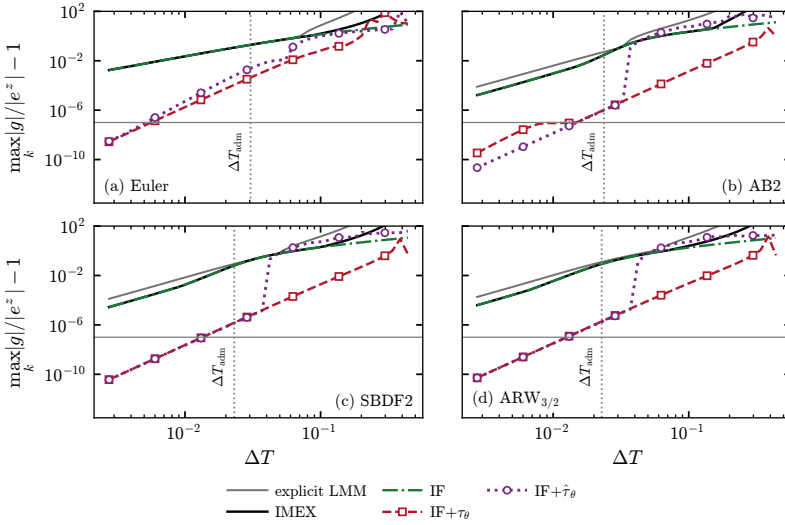


FIG. SM3.6. *Lorenz-63 system: largest flow-relative excess amplification $\max_k |g|/|e^z| - 1$ versus ΔT ; layout as in Figure 5.3 of the main paper.*

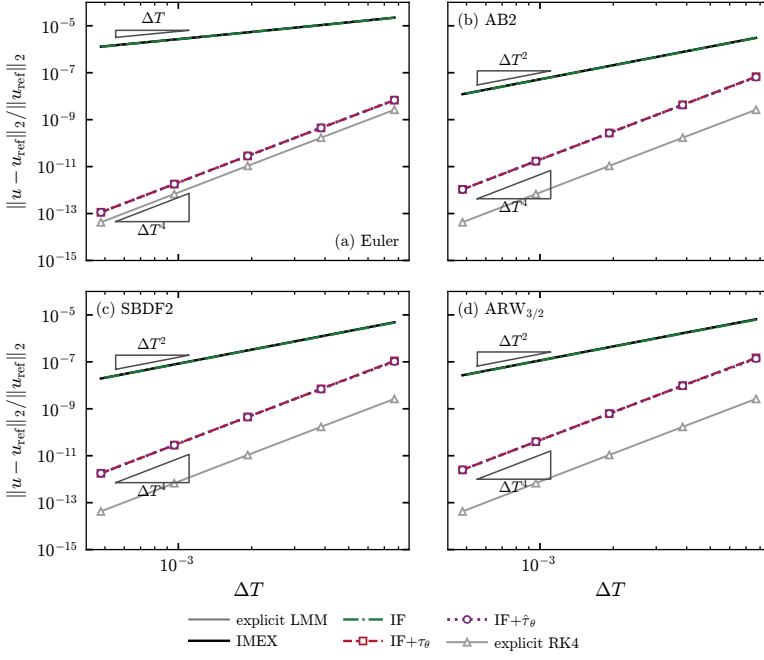


FIG. SM3.7. *Burgers testbed*: $\|u - u_{\text{ref}}\|_2 / \|u_{\text{ref}}\|_2$ at $t = 16 \Delta T_{\text{scheme}}$ versus ΔT ; layout as in Figure 5.4 of the main paper.

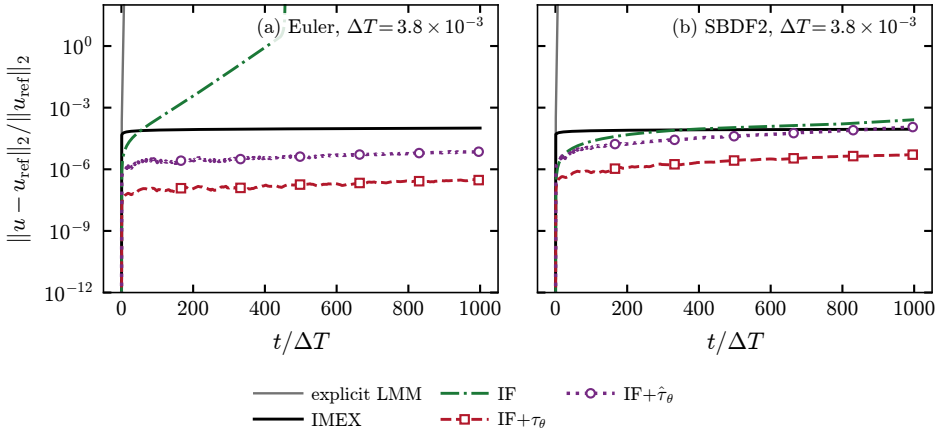
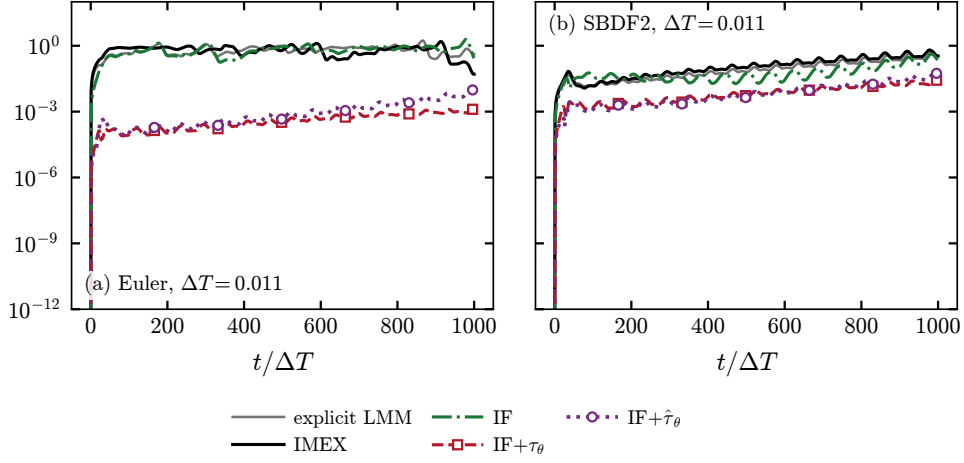
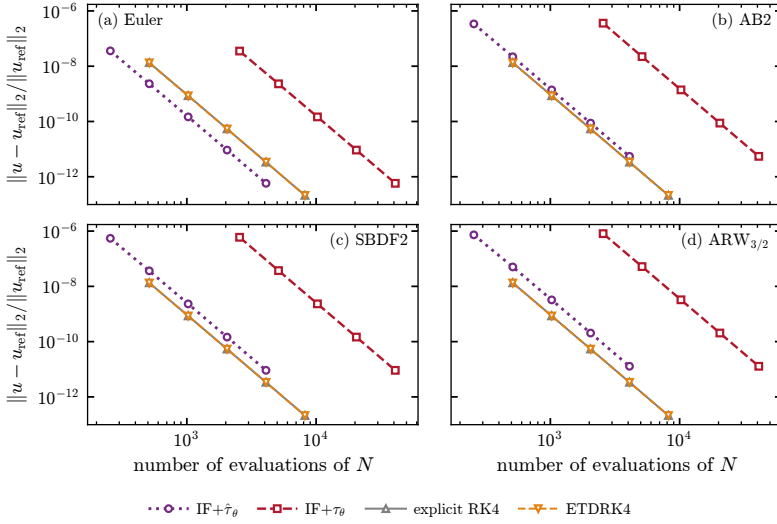


FIG. SM3.8. *KdV testbed*: $\|u - u_{\text{ref}}\|_2 / \|u_{\text{ref}}\|_2$ over 10^3 steps at $\Delta T = \Delta T_{\text{scheme}}$.


 FIG. SM3.9. *Lorenz-63 system*: $\|u - u_{\text{ref}}\|_2 / \|u_{\text{ref}}\|_2$ over 10^3 steps at $\Delta T = \Delta T_{\text{scheme}}$.

 FIG. SM3.10. *Burgers testbed*: $\|u - u_{\text{ref}}\|_2 / \|u_{\text{ref}}\|_2$ at $t = 256 \Delta T_{\text{scheme}}$ versus the total number of evaluations of N (counting as in Figure 5.8 of the main paper).

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